

Started on Wednesday, 30 April 2025, 3:08 PM**State** Finished**Completed on** Wednesday, 30 April 2025, 5:56 PM**Time taken** 2 hours 48 mins**Overdue** 48 mins 12 secs**Grade** 100.00 out of 100.00Question **1**

Correct

Mark 20.00 out of 20.00

Write a python program to traverse the elements in doubly linked list.

Answer: (penalty regime: 0 %)

Reset answer

```

1 class Node:
2     def __init__(self, data):
3         self.item = data
4         self.nref = None
5         self.pref = None
6 class DoublyLinkedList:
7     def __init__(self):
8         self.start_node = None
9     def insert_in_emptylist(self, data):
10        if self.start_node is None:
11            new_node = Node(data)
12            self.start_node = new_node
13        else:
14            print("list is not empty")
15    def insert_at_start(self, data):
16        if self.start_node is None:
17            new_node = Node(data)
18            self.start_node = new_node
19            print("node inserted")
20            return
21        new_node = Node(data)
22        new_node.nref = self.start_node

```

	Expected	Got	
✓	18	18	✓
	5	5	
	10	10	
	50	50	
	29	29	
	39	39	
	49	49	

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question 2

Correct

Mark 20.00 out of 20.00

Write a program in Python to calculate the value of the following expression by using lambda function.

The expression is -

$(x + 10) + (y + 2) * z$

For example:

Input	Result
4 3 2	24

Answer: (penalty regime: 0 %)

```

1 x=int(input())
2 y=int(input())
3 z=int(input())
4 lam=(x+10)+(y+2)*z
5 print(lam)
6
7
8

```

	Input	Expected	Got	
✓	4 3 2	24	24	✓
✓	20 3 10	80	80	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **3**

Correct

Mark 20.00 out of 20.00

Define a function to delete an element from a specific location in the given linked list.

Answer: (penalty regime: 0 %)

Reset answer

```

1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None
5 class delete_front:
6     def __init__(self):
7         self.head = None
8
9     def removeNode(self, position):
10        if self.head==None:
11            return
12        temp=self.head
13        if position==0:
14            self.head==temp.next
15            temp=None
16            return
17        for i in range(position-1):
18            temp=temp.next
19        temp1=temp.next.next
20        temp.next=temp1
21    def push(self, data):
22        if self.head is None:

```

	Input	Expected	Got	
✓	5 10 20 30 40 50	Enter the number of elements to push: 50 40 30 10	Enter the number of elements to push: 50 40 30 10	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question 4

Correct

Mark 20.00 out of 20.00

Write a python program to insert an element before the specified item in singly linked list.

Answer: (penalty regime: 0 %)

Reset answer

```

1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None
5 class LinkedList:
6     def __init__(self):
7         self.head = None
8     def traverse_list(self):
9         if self.head is None:
10            print("List has no element")
11            return
12        else:
13            n = self.head
14            while n is not None:
15                print(n.data , " ")
16                n = n.next
17    def insert_at_start(self, data):
18        new_node = Node(data)
19        new_node.next = self.head
20        self.head= new_node
21    def insert_at_end(self, data):
22        new_node = Node(data)

```

	Expected	Got	
✓	After inserting elements at the end 10 20 30 After inserting elements at the beginning 5 10 20 30 Inserting elements before the specified item 5 10 15 20 30	After inserting elements at the end 10 20 30 After inserting elements at the beginning 5 10 20 30 Inserting elements before the specified item 5 10 15 20 30	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **5**

Correct

Mark 20.00 out of 20.00

Type a python function to insert elements at the beginning of the doubly linked list.

Answer: (penalty regime: 0 %)

Reset answer

```

1 class Node:
2     def __init__(self, data):
3         self.item = data
4         self.nref = None
5         self.pref = None
6 class DoublyLinkedList:
7     def __init__(self):
8         self.start_node = None
9
10    def insert_in_emptylist(self, data):
11        if self.start_node is None:
12            new_node = Node(data)
13            self.start_node = new_node
14        else:
15            print("list is not empty")
16
17    def insert_at_start(self, data):
18        new_node = Node(data)
19        if self.start_node is None:
20            self.start_node = new_node
21        else:
22            new_node.nref = self.start_node

```

	Expected	Got	
✓	10	10	✓
	20	20	
	30	30	
	40	40	

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.